

The Basic Logic Gates



- The logic gates are the basic building blocks of a digital system. It is an electronic circuit having one or more than one Input signal and only one Output signal.
- The relationship between the input and the output is based on certain logic. At any given moment, every terminal is in one of the two conditions, either 0 or 1.
- 0=false=low
- 1=true=high
- Based on logic, the gates are named as AND , OR , NOT, NAND, NOR, EX-OR, EX-NOR

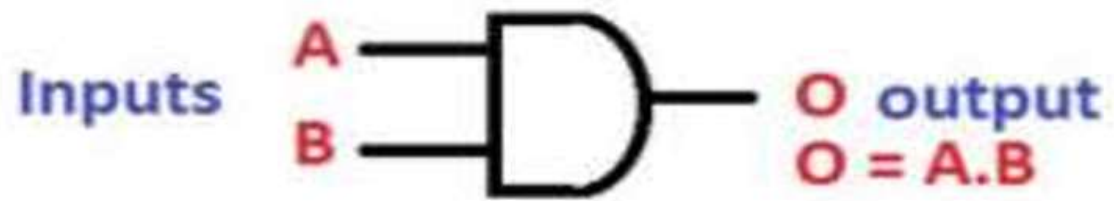
AND gate



- It has 'n' inputs and only one output.
- The output of this gate is true only when all the inputs are true.
- When one or more inputs of the AND gate inputs are false, then only the output of the AND gate is false.
- In other word,

The output is high($Y=1$) if and only if all the inputs to the gate are high (1). The output is low ($Y=0$) if one of the inputs is low (0).

AND gate



Symbol

Inputs		Output
A	B	O
0	0	0
0	1	0
1	0	0
1	1	1

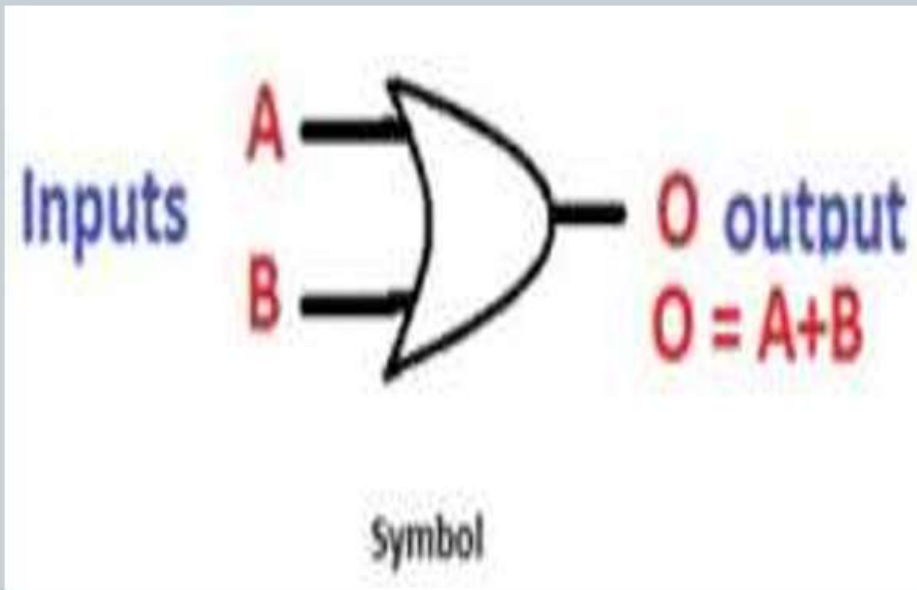
Truth table

OR gate

- The OR gate is a digital logic gate with 'n' inputs and one output.
- The output of the OR gate is true only when one or more inputs are true.
- If all the inputs of the gate are false, then only the output of the OR gate is false.
- In other word,

The output is high($Y=1$) if anyone input is high (1).
The output is low ($Y=0$) if and only if all inputs are simultaneously low (0).

OR gate



Inputs		Output
A	B	O
0	0	0
0	1	1
1	0	1
1	1	1

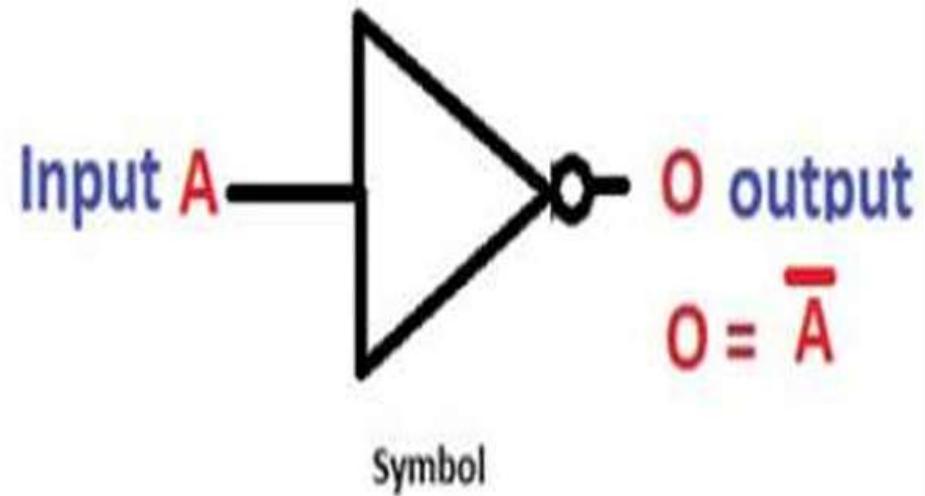
Truth table

NOT gate



- The NOT gate is a digital logic gate with only one input and only one output.
- It operates an inverter operation of the input.
- When the input of the NOT gate is true then the output will be false and vice versa.

NOT gate



Inputs	Output
A	0
0	1
1	0

Truth table

The universal Logic gate



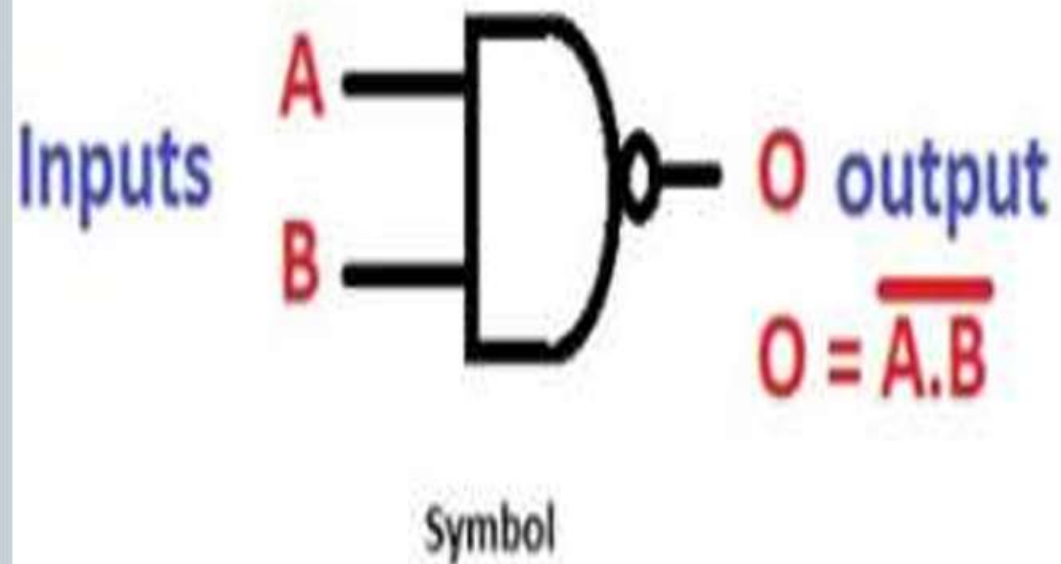
- A universal gate is a logic gate which can implement any Boolean function without any need to use other types of gates.
- The NAND & NOR Gates are called Universal gates.

NAND gate



- The NAND gate is a digital logic gate with 'n' inputs and one output.
- Performs the operation of the AND gate followed by the operation of the NOT gate.
- NAND gate is designed by combining the AND & NOT gates.
- If the both inputs of the NAND gate are high, then the output of the gate will be low otherwise high.

NAND GATE



Inputs		Output
A	B	O
0	0	1
0	1	1
1	0	1
1	1	0

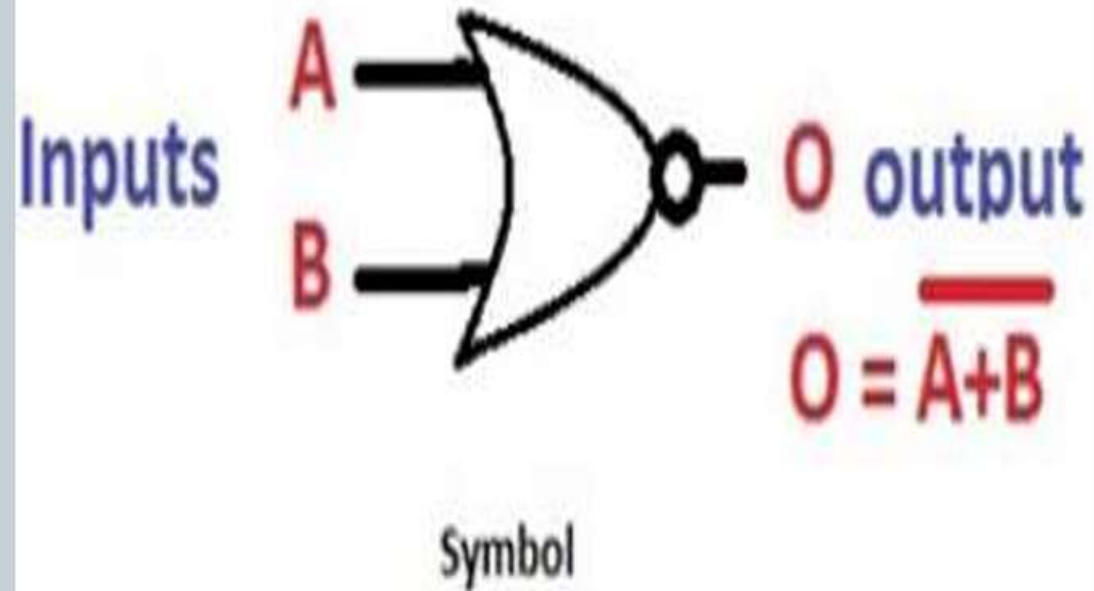
Truth table

NOR gate



- The NOR gate is a digital logic gate with 'n' inputs and one output
- performs the operation of the OR gate followed by the NOT gate.
- NOR gate is designed by combining the OR and NOT gate
- When any one of the input of the NOR gate is true(high), then the output of the NOR gate will be false(low).

NOR gate



Inputs		Output
A	B	O
0	0	1
0	1	0
1	0	0
1	1	0

Truth table

The Special purpose logic gates

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- **EX-OR**
- **EX-NOR**

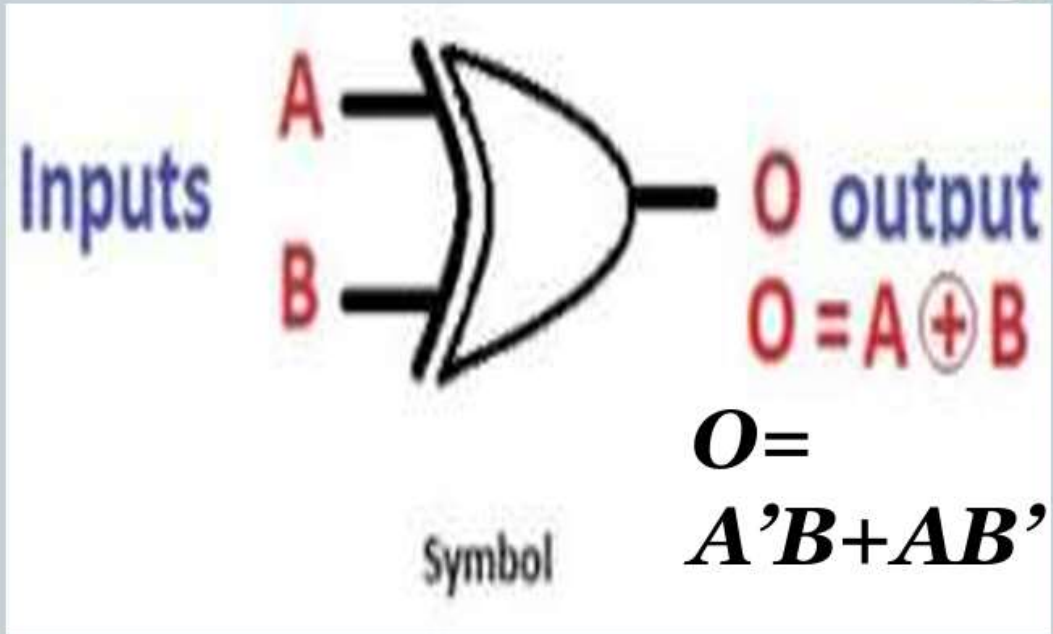
EX-OR gate (Exclusive OR)



- The Exclusive-OR gate is a digital logic gate with two inputs and one output.
- If any one of the inputs of this gate is high, then the output of the EX-OR gate will be high.
- In other word

The output is high if inputs are different and low if both inputs are same

EX-OR gate



Inputs		Output
A	B	O
0	0	0
0	1	1
1	0	1
1	1	0

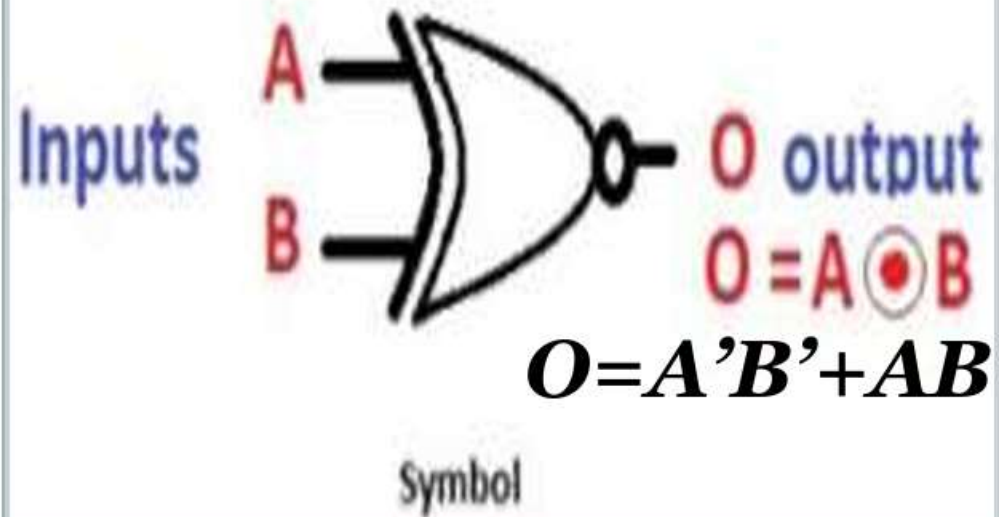
Truth table

EX-NOR gate



- The Exclusive-NOR gate is a digital logic gate with two inputs and one output.
- It performs based on the operation of NOR gate
- When both the inputs of this gate are same then the output of the EX-NOR gate will be high.
- But, if any one of the inputs is high (but not both), then the output will be low.

EX-NOR gate



Inputs		Output
A	B	O
0	0	1
0	1	0
1	0	0
1	1	1

Truth table